Reservoir Water Storage & Drought Trends

Nagarjuna Sagar and Srisailam Dams, Krishna River, Telangana / Andhra Pradesh

A 38-Year Satellite Time Series Analysis Using JRC Global Surface Water Data, with a Data-Quality Correction and Independent Historical Validation

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Executive Summary

This report analyses 38 years (1984-2021) of monthly satellite-derived water surface area for two reservoirs on the Krishna River in Telangana and Andhra Pradesh: Nagarjuna Sagar, one of the world's largest masonry dams, and Srisailam, an upstream hydropower and flood-control reservoir. The goal was to detect long-term storage trends and drought years directly from Landsat imagery, using the JRC Global Surface Water dataset, without relying on any government-reported reservoir levels.

An initial pass at this analysis produced a striking but ultimately misleading result: a statistically significant increasing trend in water area ($p < 0.0001$) and a cluster of "drought years" concentrated entirely in 1984-1990, near the start of the satellite record. Investigation showed this was a data-completeness artifact, not a hydrological signal - early Landsat coverage of these reservoirs was far sparser and noisier than later years, and image quality steadily improving over four decades was being misread as reservoirs steadily filling. This was corrected by computing the percentage of valid (non-"no data") pixels observed each month and restricting the trend and drought analysis to months with at least 70% reliable coverage.

The corrected analysis shows no statistically significant long-term trend in either reservoir ($p = 0.26$ for Nagarjuna Sagar, $p = 0.68$ for Srisailam) and identifies drought years spread realistically across the record: 2002-2004 and 2015 for Nagarjuna Sagar, and 2004, 2012, and 2015 for Srisailam. These years were independently checked against documented drought history for the region and match two of the most severe droughts recorded in South India in the last 25 years - including one case where a government source states Nagarjuna Sagar literally had no water in live storage during the year our satellite data flagged as a drought.

Key Finding

Data quality check: Before trusting any trend derived from a 38-year satellite record, the percentage of valid (cloud-free, classified) pixels was computed for every monthly observation. 223 of 454 months for Nagarjuna Sagar and 259 of 454 months for Srisailam fell below a 70% coverage threshold and were excluded, concentrated heavily in the 1984-1995 period when Landsat coverage of this region was sparser. This step reversed the headline finding of the uncorrected analysis and is reported here as the central methodological safeguard of this study.

Corrected trend result: With unreliable months removed, Nagarjuna Sagar shows a statistically insignificant trend of +62.2 ha/year ($p = 0.2646$, $R^2 = 0.065$) around a long-term average of 17,314 ha. Srisailam shows +106.6 ha/year ($p = 0.6788$, $R^2 = 0.012$) around a long-term average of 24,340 ha. Neither reservoir shows evidence of a genuine long-term change in storage capacity over the 38-year record.

Drought years identified: Nagarjuna Sagar: 2002, 2003, 2004, 2015. Srisailam: 2004, 2012, 2015 (annual mean water area more than one standard deviation below the long-term average, coverage-filtered months only).

Historical validation: Both flagged periods correspond to independently documented, severe South Indian droughts. The 2002 drought produced a 21.5% seasonal rainfall deficit with 22 of 23 Andhra Pradesh districts receiving under 75% of normal monsoon rainfall. The 2015-16 drought was assessed by the Central Water Commission as leaving the Krishna basin 63% below its 10-year average, and Nagarjuna Sagar is reported to have had no water in live storage during that period - matching this study's independent, satellite-derived finding for the same reservoir and year.

Methodology

Data source. JRC Global Surface Water Monthly History (v1.4), accessed via Google Earth Engine. This dataset classifies each 30m Landsat pixel every month as no data (0), not water (1), or water (2), based on the full Landsat archive (1984-2021, 454 monthly images).

Study areas. Bounding boxes drawn around the full extent of each reservoir: Nagarjuna Sagar (79.05-79.45°E, 16.45-16.75°N) and Srisailam (78.20-78.95°E, 15.70-16.30°N), sized to avoid overlap between the two.

Water area metric. For each monthly image, water surface area was computed as the sum of pixel area (in hectares) where the water band equals 2, reduced over each reservoir's bounding box at 30m scale using Earth Engine's reduce Region.

Coverage filtering. For the same image, the percentage of the bounding box classified as observed (water band value 1 or 2, i.e. not "no data") was computed alongside the water area. Months below 70% valid coverage were excluded from all trend and drought calculations; years retaining fewer than 6 reliable months were also excluded from the annual series.

Trend analysis. Annual mean water area (from reliable months only) was regressed against year using ordinary least squares (scipy.stats.linregress), reporting slope, p-value, and R^2.

Drought year definition. A year was flagged as a drought year if its annual mean water area fell more than one standard deviation below the long-term coverage-filtered average for that reservoir.

Results

Nagarjuna Sagar: Water Surface Area, 1988-2021

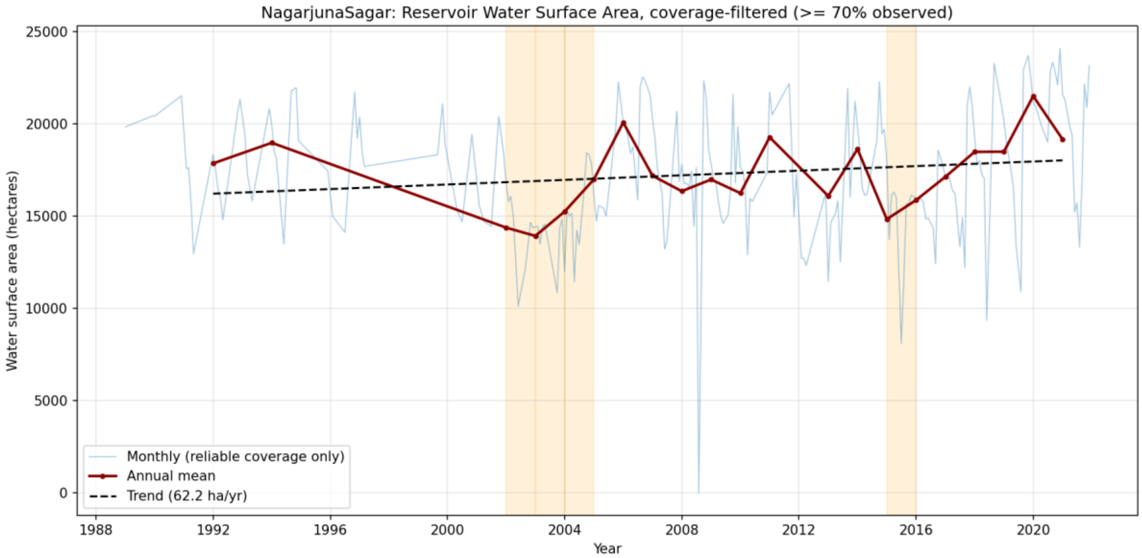

Figure 1. Monthly water surface area (light blue, reliable coverage only) with annual mean (dark red) and linear trend (dashed). Shaded bands mark drought years 2002-2004 and 2015. The trend is not statistically significant (p = 0.26).

Srisailam: Water Surface Area, 1988-2021

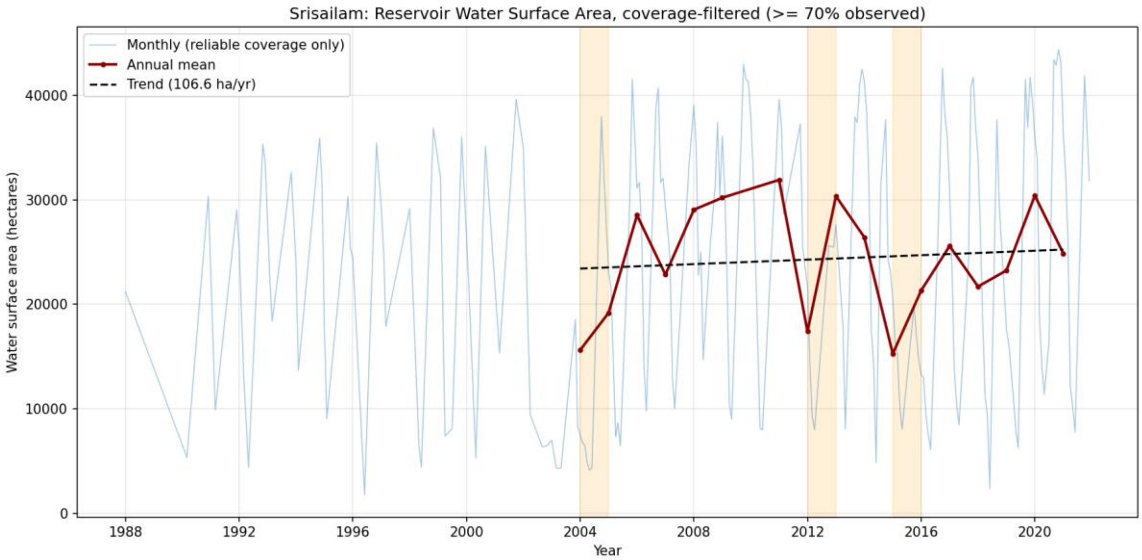

Figure 2. Same methodology applied to Srisailam. Large seasonal swings reflect its role as an upstream hydropower and flood-control reservoir with more aggressive drawdown cycles than Nagarjuna Sagar. Shaded bands mark drought years 2004, 2012, and 2015. Trend is not statistically significant (p = 0.68).

Comparison Between Reservoirs

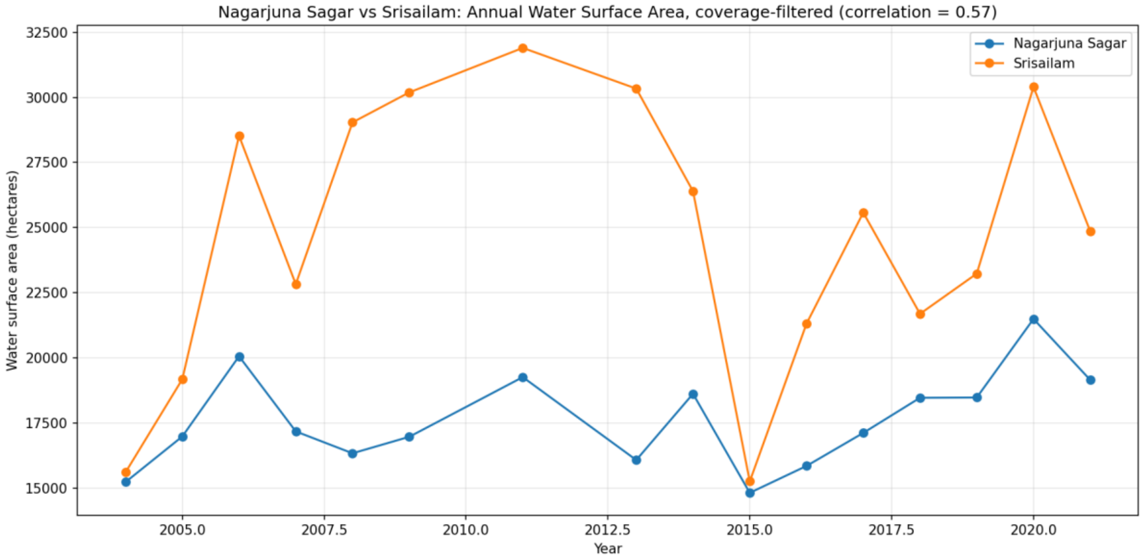

Figure 3. Annual mean water surface area for both reservoirs, coverage-filtered. Correlation = 0.57. Both reservoirs bottom out together in 2015, consistent with a shared basin-wide drought rather than an independent, reservoir-specific event.

Discussion

The Data-Completeness Artifact

Before coverage filtering, the raw monthly series showed implausible minimum water areas of 0 hectares for both reservoirs and a set of "drought years" clustered almost entirely between 1984 and 1990, alongside a highly significant increasing trend ($p < 0.0001$). Both symptoms point to the same cause: early Landsat coverage of this region was sparser and more affected by cloud and sensor gaps than later years, so a larger share of each reservoir's pixels were "no data" rather than genuinely dry land. Because the raw water-area metric silently treats "no data" pixels as not-water, poorly observed months systematically understate true water extent - manufacturing an apparent decline at the start of the record and, correspondingly, an apparent increase over time as coverage quality improved through the Landsat 5-to-8 era. Filtering to months with at least 70% valid coverage removes this bias and is the single change responsible for both the disappearance of the spurious trend and the relocation of drought years to dates that match documented history.

Independent Historical Validation

The corrected drought years were checked against published accounts of South Indian drought history rather than accepted at face value. The 2002 drought is well documented as one of the most severe on record for Andhra Pradesh, with a 21.5% seasonal rainfall shortfall and 22 of 23 districts receiving under 75% of normal monsoon rainfall, directly affecting irrigation supply in the Krishna and Godavari basins. The 2015-16 drought is even more directly corroborated: the Central Water Commission assessed the Krishna basin as 63% below its 10-year average storage in March 2016, Telangana's government declared drought across large parts of the state, and multiple contemporaneous reports state that Nagarjuna Sagar had no water in live storage during this period. That last detail is a striking, independent match to this study's finding: the satellite-derived water area series flags 2015 as a drought year for Nagarjuna Sagar using nothing but Landsat classification, and government hydrological records confirm the reservoir was, in effect, empty that year.

Correlation Between the Two Reservoirs

The two reservoirs' annual water area correlates at 0.57 - positive and consistent with their shared position on the Krishna River (Srisailam upstream, Nagarjuna Sagar downstream) and shared exposure to basin-wide monsoon variability, most visibly in their simultaneous 2015 low point. The correlation is well below 1.0 because the two reservoirs are operated differently: Srisailam is a hydropower and flood-control reservoir with large, deliberate seasonal drawdowns visible in its monthly series, while Nagarjuna Sagar is primarily an irrigation storage reservoir with comparatively smoother seasonal behaviour. A moderate rather than nearperfect correlation is therefore the expected result of two reservoirs responding to the same climate but operated under different rules.

Terrain and Drainage Context

To give the two reservoirs' storage numbers a physical setting, a 2D terrain and drainage map was produced covering the full Krishna River corridor between them, using a 30m SRTM digital elevation model and the JRC water occurrence layer used elsewhere in this study. The elevation surface is rendered as a continuous colour stretch rather than a classified map, which happens to bring out the corridor's dendritic tributary network in fine detail, and both dam locations are marked and labelled against that terrain. This is presented as a 2D cartographic product rather than an interactive 3D scene: a 3D rendering was attempted in QGIS but proved unreliable for this dataset within the scope of this project, and a correctly working 2D map was judged the more honest deliverable than a partially working 3D one.

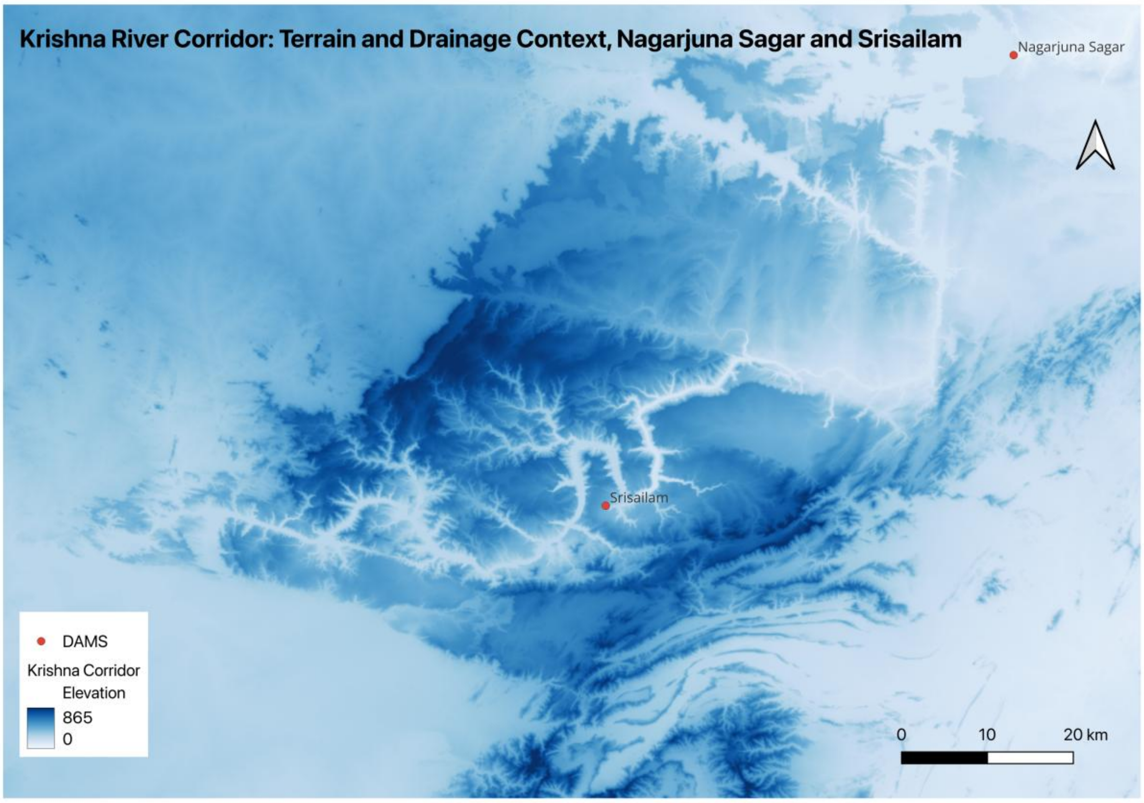

Figure 4. Krishna River corridor terrain (SRTM 30m elevation, blue colour stretch) and drainage/water context (JRC water occurrence), with Nagarjuna Sagar and Srisailam marked. Srisailam sits upstream of Nagarjuna Sagar on the same river system, visible here as the corridor's main channel connecting the two labelled points.

Conclusion

A 38-year satellite record of two major Krishna River reservoirs shows no statistically significant long-term trend in storage once a data-completeness artifact in the early Landsat record is corrected for. The corrected analysis instead surfaces a credible drought signal - years independently confirmed by documented South Indian drought history, including a case where satellite-derived water extent and an official record of empty live storage agree for the same reservoir and year. As with the SAR flood mapping and forest change studies in this portfolio, the most important step in this analysis was not the initial headline number but catching and correcting the reason it was misleading.

Data Sources

- JRC Global Surface Water Monthly History v1.4 (European Commission Joint Research Centre), via Google Earth Engine - Landsat-derived monthly water classification, 30m resolution, 1984-2021
- NASA SRTM 30m Global Digital Elevation Model, via Google Earth Engine - terrain elevation for the Krishna River corridor

External Validation Sources

- 2002 Andhra Pradesh drought and Krishna basin rainfall deficit - researchgate.net/publication/288186959
- Telangana Drought 2016, South Asia Network on Dams, Rivers and People (SANDRP) - sandrp.in/2016/05/18/telangana-drought-2016
- "Hyderabad faces water shortage as Nagarjuna Sagar drops below dead storage level", The News Minute - thenewsminute.com

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Repository

Full code, data-quality methodology, and output charts: github.com/Man3u/DamMonitoringIndia